Three layers, two loops: what is frozen, what learns, and why they are separated

Model weights frozen always; harness structure frozen per generation and learned by search in the slow loop; the decisions inside the harness learned by gradients in the fast loop

THE STACK

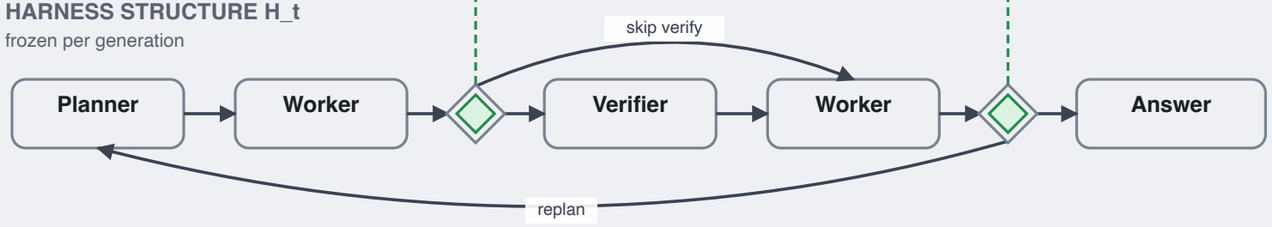
CONTROLLER π_θ
learns

π_θ: the decisions (tier, delegate, verify, replan, stop)
factored out of the code into a few thousand parameters
18 priced actions over 88 features; updated by the fast loop only

in an ordinary harness these diamonds are `if` statements and config fields inside the grey band — that is what DarwinX evolves

factored out

HARNESS STRUCTURE H_t
frozen per generation



roles, tools, prompts, SOQL path, 8-call cap: written as code and config changes only when a new generation ships (DarwinX stage, our scripted stages)

MODEL WEIGHTS
frozen always

Haiku 4.5 | Sonnet 5.5 | Opus 5.5 | open-weight agent models

the API tiers and any open-weight agent models; every role call above runs on one of them
nothing in PACE updates them

their layer (SFR-RL, RISE, PMD → weights): the slow-slow loop, hours–days, not drawn

THE TWO LOOPS

FAST LOOP · minutes · every request — touches only the green band

θ

serve with ε
incumbent π_θ on live traffic; propensities logged

log
(features, action, propensity, cost, outcome)

train candidate by gradients
cGRPO / AWR+IPS; λ by dual ascent

DR GATE
quality lower CI ≥ q_prev – ε
AND cost upper CI ≤ B

promote / revert
candidate serves tomorrow, or the incumbent keeps serving

THE COUPLING QUESTION
a failure or drift event goes to which loop? edit the structure · adapt the controller · do nothing — decided per event, under cost
continual study: most drift absorbed by the fast loop; stage changes need the slow one

SLOW LOOP · days · per generation — touches only the middle band

propose structure edit
evolver / engineer

run on the benchmark
hundreds of episodes per candidate

GATE
avg@k AND priced-BWT ≥ threshold

ship H_{t+1}
the next generation of the grey band

edits H_t only

WHY THE LAYERS ARE SEPARATED (not only because it is hard)

	structure (slow loop)	decisions (fast loop)
signal	discrete code edit; learned by search (propose → benchmark → keep/discard); hundreds of episodes per candidate	parameter choice; learned by gradients; one update per request
validation	a benchmark run — no logged counterfactual exists for a structure that never served	an off-policy estimate on last week's logs (IPS / SNIPS / DR with intervals)
blast radius	where forgetting lives: –34.7% BWT from adding specialists alone (EvoHarnessBench 2609.04280)	bounded by the gate's interval; the structure is unchanged
metrics	BWT / FWT / regret need the same structure to compare	defined on a fixed structure; computed every day

NEXT*: the hierarchical controller (proposed, not run)

a green diamond one level up choosing among 2–3 pre-built structures per request: structure SELECTION becomes a logged, gated decision (options; SkillOrchestra's routing made learnable)
the fully learned structure per request (RAO 2605.06639, Meta-Harness 2603.28052) is unpriced and ungated and ~100× the episodes per unit of learning — a worse trade, not only a harder one

LEGEND frozen | learns | gate | dashed / * = proposed, not run | thin green = fast loop | thick grey = slow loop

'frozen' on the middle band means unchanged between generations, not unchangeable

Sources: PACE README; results/mas_continual/RESULTS.md; prep_course §9 and §14. * = proposed, not run.